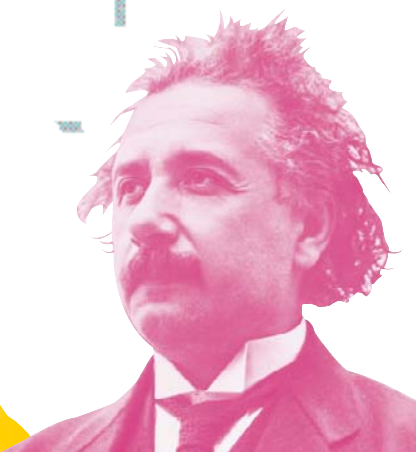


I can calculate
the motion of
heavenly bodies
but not the
madness of people

God does
arithmetic

If at first an idea
is not absurd,
then there is no
hope for it



NEWTON
1643–1727

Inspired by Galileo's study of falling objects and Kepler's elliptical orbits, Isaac Newton worked out how gravity holds the Universe together. He explained gravity like this. Throw a stone sideways and it falls to Earth. Throw it harder and it still falls. If you could throw it hard enough, it would keep going without falling – and that's just what the Moon is doing.



EULER
1707–1783

The Swiss mathematician Leonhard Euler ("oiler") was the most prolific mathematician ever. He wrote over 800 papers, many after he had gone blind in 1766. After he died it took 35 years to publish them all. He is most famous for solving the Königsberg Bridge puzzle, which was the start of network theory – without which today's microchips could not be made.



GAUSS
1777–1855

Classed as the third greatest mathematician in history (after Archimedes and Newton), Karl Gauss was correcting his father's sums when he was 3. As a schoolboy he found an ingenious way of adding consecutive numbers quickly. Gauss also proved that any number is the product of primes, in one way only ($8 = 2 \times 2 \times 2$; $6 = 2 \times 3$; and so on).

Gauss amused himself by *keeping records* of the lengths of famous men's lives – in days.

EINSTEIN
1879–1955

Albert Einstein realized light moves at constant speed and is pulled by gravity. He also realized that mass and energy are versions of the same thing and devised an equation to show it (below). The equation shows that a tiny amount of mass (m) equals a vast amount of energy (E), since to make them equal you have to multiply the m by a very large number: the speed of light squared (c^2).

$$E = mc^2$$